

Management and Valorization of Reverse Osmosis Reject in Hemodialysis

Feasibility Study and Sizing of a Reject Water Reuse System—Case Study
of a Hemodialysis Center in Agadir

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Abstract

Reverse osmosis, an essential step in producing ultra-pure water for hemodialysis, structurally rejects a significant fraction of the water it processes. Currently discharged without valorization in most dialysis centers, this reject represents a clean and continuously available resource. This document addresses its **management**: how can a hemodialysis center best valorize this reject – through direct water reuse, partial recycling, or recovery of a fraction of its energy – rather than simply discarding it? A comparative study of several management pathways is presented, followed by the complete technical and financial sizing of the selected solution.

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1 Center Overview

The study is conducted within a medium-sized hemodialysis center, a specialized healthcare facility dedicated to the management of patients with end-stage chronic renal failure. The center operates continuously, with sessions organized in multiple daily rotations (typically a morning session and an evening session, each lasting approximately 4 hours) to accommodate a large number of patients relative to its capacity.

A multidisciplinary team (nephrologists, specialized nurses, dialysis technicians) ensures the medical care and daily technical monitoring of the installations, particularly the water treatment room, whose proper operation directly dictates the safety of dialysis sessions.

Like any center of this type, the facility features a dedicated water treatment room (filtration, softening, activated carbon, reverse osmosis) that supplies ultra-pure water to the dialysis stations, as well as ancillary operational needs (housekeeping, laundry, patient catering, air conditioning) which have historically been handled independently of the medical water production.

2 Reverse Osmosis in Hemodialysis

2.1 Role of Water in Hemodialysis

A hemodialysis session relies on continuous mass exchange between the patient's blood and a dialysis fluid (dialysat) across a semi-permeable membrane. This dialysat is prepared using water that meets extremely strict purity requirements – classified as a drug by the Pharmacopoeia – as it remains in indirect continuous contact with the patient's blood throughout the treatment.

To achieve this level of purity, municipal water undergoes a multi-stage treatment train: coarse filtration, softening (removal of calcium and magnesium), activated carbon filtration (removal of chlorine and organic compounds), and finally reverse osmosis. This last step is the most demanding: water is forced under pressure through a membrane whose pores only let water molecules pass while retaining salts, bacteria, endotoxins, and viruses.

2.2 RO Reject: A Neglected Resource

The physical principle of reverse osmosis dictates that a portion of the water fed into the system never reaches the other side of the membrane; it emerges concentrated with unwanted elements in the form of **reject** (or concentrate). On the studied installation, the main operating data were **recorded directly on-site during the internship period**. The measured average reject flow rate is **2,400 L/h**, representing **57,600 L/day** when extrapolated over 24 hours. This flow rate accounts for approximately **34% of the incoming water flow rate** under the considered operating conditions.

This reject does not originate from the blood circuit and is not in contact with blood or dialysat. It is pre-treated water resulting from the reverse osmosis process, but its composition differs from that of water intended for dialysis. Its reuse must therefore be limited to applications compatible with its quality and be subject to appropriate sanitary and technical verifications.

2.3 Problem Statement

In a country facing growing water stress, the broader question raised by this work is: **how best to manage this resource?** Three families of answers can be envisioned: direct water reuse for non-medical purposes, partial recycling within the process itself, and recovery of a fraction of its residual energy. This document systematically evaluates these pathways before proposing a complete technical and financial solution.

2.4 Baseline Data

Average reject flow rate	2,400 L/h (0.667 L/s) – measured on-site
Daily available volume	57,600 L/day – 24h extrapolation
RO feed pressure	2 to 10 bar – recorded on the installation
Total electrical power demand	~27.5 kW continuous – sizing data

Measurement Methodology. The main flow and operational data used in this study were recorded directly on the installation during the internship period. Measured values are distinguished from extrapolations, sizing assumptions, and economic estimates.

3 Comparative Study of Reject Management Solutions

Before selecting a solution, every conceivable pathway was evaluated, calculated, and confronted with the physical and economic reality of a dialysis center. This section presents the calculations that led to discarding certain energy pathways, retaining one, and evaluating direct water valorization pathways.

3.1 Pathway 1 – Closed-Loop Hydroelectric Micro-Plant

Principle. A reservoir feeds a turbine; the turbinized water is then pumped back to the reservoir to recirculate continuously (loop flow rate $Q = 1.8 \text{ m}^3/\text{s}$, drop height $h = 2.3 \text{ m}$).

$$P_h = Q \times \rho \times g \times h$$

Numerical application – hydraulic power on the turbine side:

$$P_h = 1.8 \times 1000 \times 9.81 \times 2.3 = 40,613 \text{ W} \approx 40.6 \text{ kW}$$

Electrical power produced (turbine 80%, generator 95%, combined efficiency 76%): $P_{elec} = 40.6 \times 0.76 \approx 30.9 \text{ kW}$.

Power required by the pump to lift the same flow rate to the same height, *even with a theoretically perfect pump* (100% efficiency, unachievable in practice):

$$P_{pump}^{ideal} = P_h / \eta_{pump} = 40.6 / 1.00 = 40.6 \text{ kW}$$

Assessment, even in the best theoretical case: $30.9 - 40.6 = -9.7 \text{ kW}$. With a realistic pump efficiency (60–75%), the deficit reaches -23.3 to -36.8 kW .

Verdict: Discarded. This configuration violates energy conservation and cannot produce a net gain, regardless of the chosen components.

3.2 Pathway 2 – Reject Only, Single Gravity Pass

Principle. The actual reject flow rate, without a recirculation pump, passes through a turbine once before standard disposal.

$$P_{elec} = Q \times \rho \times g \times h \times \eta_{turbine} \times \eta_{generator}$$

Drop Height	Electrical Power	Share of Demand (27.5 kW)
2 m	9.9 W	0.036%
5 m	24.9 W	0.090%
10 m	49.7 W	0.181%
20 m	99.4 W	0.361%
60 m	298.2 W	1.084%
100 m	497.0 W	1.807%
200 m	994.1 W	3.615%

Verdict: Discarded. Even at a 200 m drop height, the obtained power covers only 3.6% of the demand: the available flow rate is structurally insufficient, independent of height.

3.3 Pathway 3 – Pooling Between Dialysis Centers

Principle. Collecting the reject from multiple RO units across dialysis centers in the same urban area toward a common turbine (assumption: 6 centers, 2 L/s per RO unit, combined flow 12 L/s).

Height	Power (6 centers)
10 m	0.89 kW
20 m	1.79 kW

Verdict: Not selected. At last, an order of magnitude worth noting – but this would require a collection network connecting geographically separated clinics, which falls outside the scope of a single center. Documented here as a future perspective.

3.4 Pathway 4 – Pressure Energy Recovery on the Reject Line

Principle. Unlike previous pathways, this solution exploits the residual pressure of the reject at the RO outlet – pressure already generated by the existing feed pump and normally wasted through the circuit's pressure restrictors.

$$P_h = Q \times \Delta P$$

Destroyed Pressure	Theoretical Power (85% efficiency)
2 bar	113 W
3 bar	170 W
5 bar	283 W
8 bar	453 W

Sizing note: this calculation assumes an industrial pressure exchanger (85% efficiency), representative of devices used in large-scale desalination. Commercially available components at this power and flow rate – DC hydroelectric micro-generators – have a much more modest manufacturer nominal power, around 10 W per unit (see actual sizing, section 4.4).

Verdict: Selected, to power the water reuse subsystem – requiring no drop height, civil engineering, or additional pump.

3.5 Pathway 5 – Direct Reuse of Reject Water

Usage	Estimated Demand	Reliability
Housekeeping (cleaning, toilets, handwashing)	2,000 L/day	Estimate
Washing machine + kitchen (linens, meals)	2,040 L/day	Estimate
Cooling tower makeup	1,100 L/day	Estimate
Backwash / softener and filter regeneration	150–400 L/day	Industry data
Partial recycling upstream of RO (conductivity sensor)	variable	Documented practice since 2007 [3]
Green space watering (limited area)	400–800 L/day	Estimate (ref. Saveh, Iran)
Total Internal Uses	~6,015 L/day	Out of 57,600 L/day available (10.4%)

Verdict: Selected. The unconsumed surplus (~51,600 L/day) remains normally discharged – no additional internal uses identified to absorb it entirely.

3.6 Comparative Synthesis

Pathway	Verdict	Key Result
1. Closed-loop turbine + pump	Discarded	Negative balance even with ideal pump (−9.7 kW)
2. Reject only, gravity	Discarded	< 3.6% of demand, even at 200 m
3. Inter-center pooling	Not selected	0.9–1.8 kW; heavy infrastructure required
4. Pressure recovery	Selected	~20 W actual (commercial components)
5. Water reuse	Selected	~6,015 L/day out of 57,600 L/day available

4 Selected Solution – Technical Study

4.1 General Architecture

The solution combines a water reuse subsystem and an energy recovery subsystem, both supplied by the same RO reject, but physically separated from the existing ultra-pure water circuit supplying the dialysis stations (Figure 1).

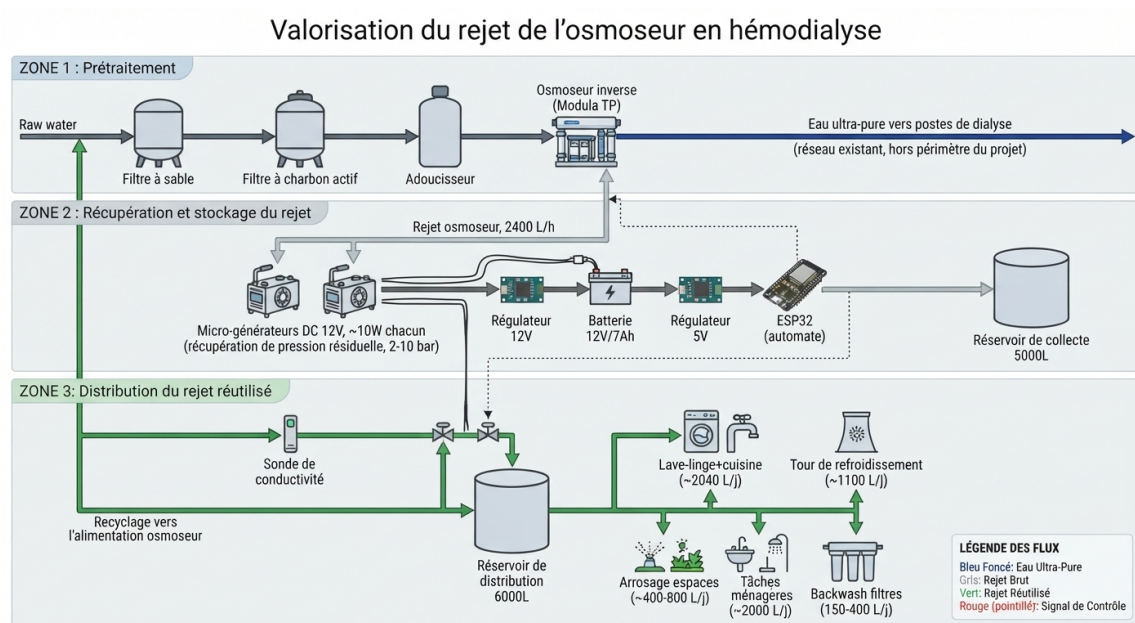


Figure 1: General architecture of the selected system.

4.2 Water Reuse Subsystem

A collection tank (5,000 L) receives the RO reject. After withdrawal for upstream recycling, the surplus feeds a distribution tank (6,000 L) supplying the uses listed in section 3.5.

4.3 Partial Recycling Upstream of the RO Unit

A fraction of the reject is proposed to be reinjected into the RO feed water, upstream of the softeners. A conductivity sensor continuously monitors the mixture's conductivity; beyond a threshold defined during commissioning, recycling is interrupted and the feed automatically switches back to municipal water alone.

This architecture constitutes a sizing proposal and must be experimentally validated on the actual installation prior to any implementation, particularly regarding water quality, RO manufacturer limits, and applicable requirements for hemodialysis water. This practice has been routinely documented since 2007 in home dialysis setups [3].

4.4 Energy Recovery Subsystem and Autonomy

Two 12V DC hydroelectric micro-generators (~10 W nominal each, allowable pressure up to 1.2 MPa) are proposed in parallel on the reject line. Their actually recoverable power must be confirmed by the flow-pressure characteristics of the selected model and an on-site experimental test.

Consumption Point	Power
Controller (continuous)	5 W
7 level sensors (continuous)	3.5 W
Base consumption (continuous)	8.5 W
3 solenoid valves held open (peak)	18 W
Peak consumption	26.5 W
Available Power (2 x 10 W)	20 W

The available power covers base consumption with a +11.5 W margin; peaks are absorbed by a buffer battery.

$$\text{Autonomy} = \text{Capacity (Wh)} / \text{Power Demand (W)}$$

Selected battery: AGM 12V / 7Ah (84 Wh). Autonomy under base load: $84/8.5 \approx 9.9$ h. Autonomy under sustained peak: $84/26.5 \approx 3.2$ h – well above the actual duration of a valve opening cycle.

Value of this subsystem: operational resilience gain – control of the water reuse system remains active during general grid power outages, which is relevant in a healthcare center operating continuously. It is not a significant financial savings source (section 5).

4.5 Risk Analysis and Reliability (FMEA)

For a project applied in a healthcare environment, equipment reliability and sanitary safety are absolute imperatives. Although RO reject is not in direct contact with patients' blood, its use for sensitive purposes (handwashing, catering, or upstream recycling) is subject to strict hospital hygiene regulations.

Consequently, intended uses must be validated by the facility's hygiene officer, and supplementary treatment (such as UV disinfection or additional filtration) may be required prior to any use involving potential contact with staff or patients.

Furthermore, to anticipate potential control system failures and ensure operational safety, a Failure Mode and Effects Analysis (simplified FMEA) was conducted:

Component	Failure Mode	Probable Cause	Consequence	Preventive / Corrective Action
Conductivity sensor	Measurement drift or failure	Fouling, sensor wear	Risk of injecting inadequate water into upstream recycling	Secure automatic switch to municipal water alone (degraded mode) + alert
Micro-generators	Turbine blockage / leak	Suspended particles in reject	Loss of system electrical autonomy	Installation of a strainer filter upstream of recovery line
Buffer battery (12V/7Ah)	Charge loss / end of life	Aging, prolonged outage	Control shutdown and solenoid valve closure	Preventive maintenance (triennial replacement) and monitoring by ESP32
Solenoid valves (SV)	Stuck open or closed	Scaling, over-pressure	Tank overflow or supply disruption	Selection of robust industrial valves + safety level sensors

Table 1: Summary table of the risk analysis (FMEA).

Fail-Safe Principle: The system is designed such that in the event of a sensor, valve, or power source failure, the system defaults to a safe configuration (closing reuse valves and relying exclusively on the municipal water supply).

5 Hardware Selection and References

Component	Selected Reference	Characteristics
Hydroelectric micro-generator	Hydro DC micro-generator (generic cat.)	12 V DC, ~ 10 W, P_{max} 1.2 MPa
Transfer pump	Pedrollo PKm 60	0.5 HP, $Q_{max} \sim 40$ L/min, Head ~ 30 m
Solenoid valves	12V DC 1/2" PN16 NC Solenoid valve	12 V DC, 0.02–1.6 MPa
Level sensors	Vertical float switch ON-OFF	Discrete binary output
Flow meters	Hall effect sensor YF-S201	DN15–20, ~ 1 –30 L/min
Conductivity sensor	Analog EC sensor (Gravity/DFRobot)	0–20 mS/cm, 0–3.3 V output
Buffer battery	Sealed AGM 12V/7Ah	84 Wh
Controller	ESP32 DevKit V1 (30 pins)	Wi-Fi/BLE, ~ 80 –160 mA active
Relay module	4-channel 5V opto-isolated relay module	12V solenoid valve control

5.1 Automation

An ESP32 module is selected for the control prototype rather than a classical industrial PLC (Siemens LOGO!, Schneider Zelio), primarily due to its low power consumption, cost, and compatibility with potential MQTT/Node-RED supervision. For commissioning in a healthcare environment, the selection of appropriate industrial equipment and safety requirement analysis must nevertheless be studied separately.

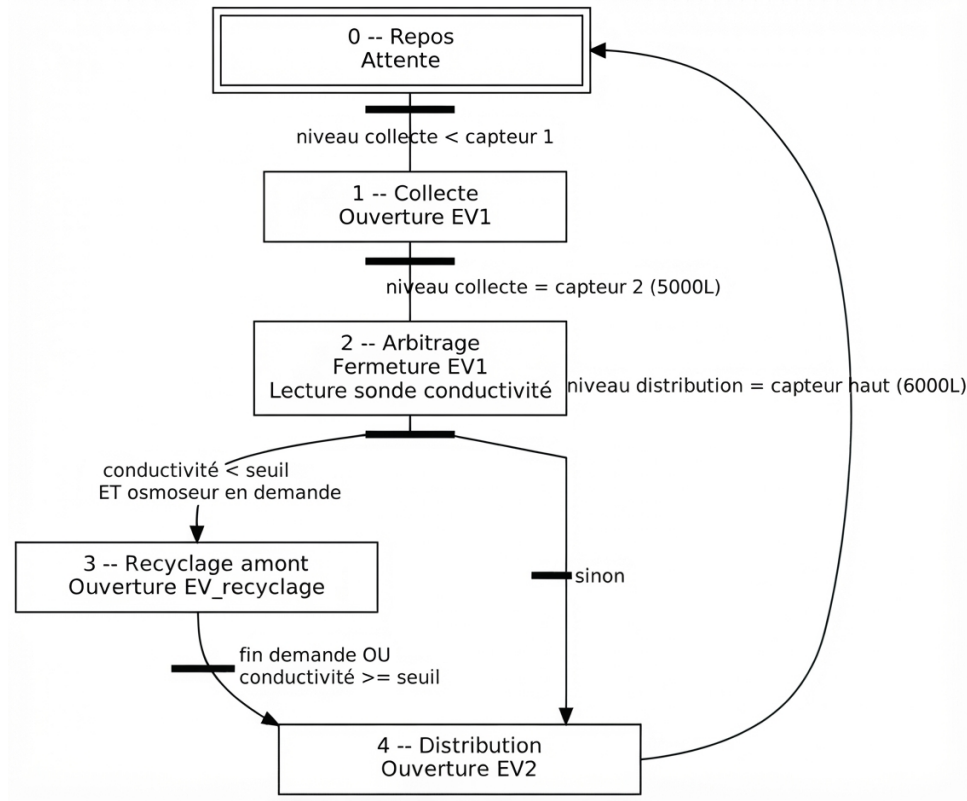


Figure 2: Grafcet of the control logic.

Control Logic (pseudocode):

```

IF collection_level < low_threshold THEN open(SV1)
IF collection_level >= high_threshold THEN
  close(SV1)
  IF conductivity < max_threshold THEN open(SV_recycling)
  ELSE open(SV2)
IF battery_voltage < critical_threshold THEN lock(all_valves)
  
```

The detailed electrical schematic (AC power circuit and DC control circuit, with legend and title block) is provided in Appendix A.

6 Financial Study

Equipment	Qty	Price (MAD)	Description
Tanks (collection 5000L + distribution 6000L)	2	15,000	Food-grade polyethylene tanks
Transfer pump (Pedrollo PKm 60)	1	5,000	Low pressure
Piping and fittings (PVC/HDPE)	1	6,000	Collection + distribution
Watering network	1	2,000	Green spaces ($\sim 75 \text{ m}^2$)
Solenoid valves	6	2,800	Distribution, recycling, collection, backup
Level sensors	7	3,500	ON-OFF float switches
Flow meters (YF-S201)	2	4,000	Collection + distribution monitoring
Conductivity sensor + controller	1	2,500	Upstream recycling control
12V DC hydroelectric micro-generators	2	800	Reject line
Buffer battery 12V 7Ah	1	800	Controller/valve autonomy
Charge controller + rectifier	1	1,000	Recovered energy conditioning
Controller (ESP32 + relays + enclosure)	1	700	Low-level control
Installation (labor)	1	3,000	Plumbing + electrical
TOTAL		47,100	

Item	Value
Reused water	$\sim 2,195 \text{ m}^3/\text{year}$
Water savings (estimated tariff 10–15 MAD/ m^3 , to be confirmed)	22,000 to 33,000 MAD/year
Electricity savings (autonomous subsystem)	$\sim 97 \text{ MAD/year}$
Indicative Return on Investment (ROI)	1.4 to 2.1 years

7 Conclusion

RO reject management does not find its best valorization in electricity generation at the scale of a dialysis center. The calculations performed show that the hydroelectric configurations studied from the available flow rate provide limited power, while pressure recovery can be considered to power system auxiliaries.

The proposed solution thus combines water reuse for non-medical purposes, partial recycling under conductivity control, and recovery of low power to contribute to control autonomy. However, the investment cost and return on investment presented remain indicative, as they depend notably on the actual water tariff, component purchase prices, and validation of hydraulic operation.

The project thus constitutes a feasibility and sizing study, to be complemented by an experimental validation phase prior to any implementation in a healthcare facility. This approach aligns with the broader reflection surrounding "green dialysis", which aims notably to reduce the water and energy footprint of hemodialysis.

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Appendix A – Detailed Electrical Plan

